# Autonomous Search and Multi-Objective Assessment of Theoretical Wormhole Geometries

A reproducibly parameterized research prototype using paceval and Trusted Interval Computation (TINC) | Technical preliminary research report | 30 October 2026 | Not independently peer-reviewed

## Plain-language summary

A computer program autonomously generated and examined mathematical models of wormholes. Wormholes are hypothetical connections within spacetime. In most classical models of traversable wormholes, unusual matter or energy properties would be required to keep the throat open.

For one automatically generated model, a numerical measure of violation of a fundamental energy condition is approximately 89.5% smaller than for the particular classical Ellis reference studied here. This may indicate a more favorable radial distribution of energy-condition violations within the defined model and metric. It does not establish that 89.5% less exotic matter would actually be needed, nor does it demonstrate superiority over all known wormhole solutions.

The candidate also exhibits greater static tidal curvature. Consequently, the research considers several competing objectives rather than seeking a single optimum. The paceval engine evaluates the generated mathematical expressions; TINC provides numerical bounds at selected input points. No physically existing wormhole has been detected.

## Scientific abstract

Objective. To investigate autonomous symbolic search for geometrically admissible candidates within the static, spherically symmetric Morris–Thorne framework. Two objectives are assessed separately: a precisely defined negative radial null-energy-condition (NEC) integral and the maximum sampled static orthonormal tidal-curvature magnitude.

Methods. A C++17 application generates expression trees, computes symbolic derivatives and Einstein/curvature quantities, and evaluates reusable paceval computation graphs with batch calls. A nondominated candidate subset is identified. TINC checks apply to individual radius values; the integrated objectives are not interval-certified.

Results. In the recorded run (random seed 20261029; 100 candidate evaluations), the selected candidate yields an NEC objective of 0.1035396212, versus 0.9900118818 for the Ellis special case. Its maximum sampled static radial tidal-curvature component is 1.9935043451. From 100 archived candidates, 24 were reassessed on a fine grid; 19 were nondominated within that subset.

Scope. The outcome indicates a substantial improvement over selected references under the particular metric and numerical protocol employed. It does not establish global optimality, a literature-wide minimum exotic-matter requirement, complete continuum validation, or physical realizability.

## 1. Research question and physical framework

The Morris–Thorne ansatz describes a static, spherically symmetric spacetime through a shape function b(r) and redshift function Φ(r). Geometrized units G = c = 1 are used. The throat radius is normalized to $r_0 = 1$, and documented numerical comparisons extend to r = 100. This normalization is a mathematical convention, not an astronomical scale [1].

$$ds^2 = -\exp(2\Phi(r))\,dt^2 + dr^2/(1 - b(r)/r) + r^2(d\theta^2 + \sin^2\theta\,d\varphi^2)$$

At the throat b(1) = 1. The flare-out condition requires b′(1) < 1. In the modeled exterior region, 1 − b(r)/r must be positive. A finite redshift function prevents a horizon arising from a divergent redshift in this ansatz. These are necessary geometric checks, not a complete demonstration of a stable, physically traversable solution.

## 2. Representation of the autonomously generated geometry

The program generates expressions U(q) and V(q) and derives b(r) and Φ(r) from them. The candidate below corresponds to the recorded Stage-8 run. The compact expression for U is algebraically simplified from the generated expression; no subsequent parameter optimization was applied.

$$q(r) = (r - 1)/r$$

Text expression: q=(r-1)/r

$$U(q) = q + q^3 / [(1 + q^2)(1 + \cos^2(a))]$$

Text expression: U=q+(q*q*q)/((1+q*q)*(1+cos(a)*cos(a)))

$$a = 1.7573959995777182$$

Text expression: a=1.7573959995777182

$$b(r) = [1 + q(r)\, U(q(r))]/r$$

Text expression: b=(1+q*U)/r

$$\Phi(r) = -1.9935043450750545/r$$

Text expression: Phi=(0-1.9935043450750545)/r

The text expressions are provided for readability. In actual paceval evaluation, auxiliary definitions are expanded into complete mathematical expressions; symbolic derivatives and physical diagnostic quantities are then constructed from those expressions.

## 3. Numerical methods: paceval and TINC

### 3.1 Reusable mathematical computation graphs

paceval is a mathematical software engine that parses textual expressions once and transforms them into an internal execution structure. The prepared structure can subsequently be evaluated repeatedly for new inputs. The application uses, among other interfaces, paceval_dGetComputationResultExt to evaluate expressions at batches of radii. Object reuse and batch processing can reduce repeated-call overhead; this alone does not establish a performance advantage over independently benchmarked alternative libraries.

### 3.2 Meaning and limitations of Trusted Interval Computation

In the paceval mode examined here, Trusted Interval Computation (TINC) provides numerical lower and upper bounds for specified input values. The program uses these results, for example, to classify the sign of a radial NEC quantity at selected radii as negative, nonnegative or inconclusive. Such classification relies on the meaning of the bounds and error codes as specified by the SDK. Independent formal verification of the TINC implementation is outside the scope of this report.

Importantly, a successful TINC result at r = 1.25 does not establish any statement about every radius between r = 1.20 and r = 1.30. The present code does not implement validated interval evaluation over continuous input domains. Consequently, the numerically computed maxima and integrals do not carry proven global error bounds.

### 3.3 Exact definition of the negative NEC objective

The principal objective is not a directly measurable amount of required matter. The code denotes the radial quantity by NEC8 = $8\pi(\rho + p_r)$. The objective is numerically integrated using the trapezoidal rule:

$$J_NEC = \int_1^{100} \tfrac{1}{2}\, r^2 \max(0, -NEC8(r))\, dr$$

This nonnegative, normalization-dependent radial integral weights the negative part of NEC8. It is neither a time-integrated averaged null energy condition (ANEC) along complete null geodesics nor automatically the “volume integral quantifier” defined by Visser, Kar and Dadhich [2]. The relative reduction stated below is meaningful only for the same explicit objective definition and evaluation protocol.

## 4. Reference models and scope of comparison

The numerical benchmarks are deliberately limited. Morris–Thorne specifies a general metric family, not a unique reference solution. The massless Ellis special case is a recognized static baseline [3]. Two constant-U geometries are constructed benchmarks for this study; they are not attributed as original solutions to any cited author.

| Model | Shape function or U(q) | Redshift Φ(r) |
|---|---|---|
| Ellis special case | b(r) = 1/r; U = 0 | 0 |
| Ellis-type form, modified redshift | b(r) = 1/r; U = 0 | −0.75/r |
| Constructed reference A | U = 1.90 | 0 |
| Constructed reference B | U = 1.99 | 0 |
| Autonomous candidate | U = q + q³/[(1+q²)(1+cos²a)] | −1.993504345/r |

This is not a comparison of every known wormhole family. Rotating, dynamical and thin-shell geometries, as well as models based on modified gravitational theories, have different mathematical assumptions and cannot be placed in the same quantitative ranking without consistently specified matter models, boundary conditions and objective functions.

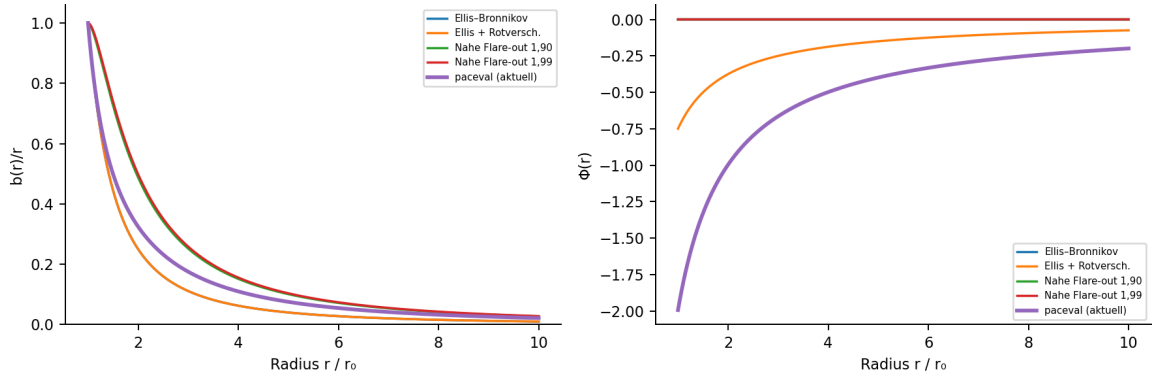



Figure 1. Modeled radial profiles of the five specifically evaluated geometries; these are not observational data.

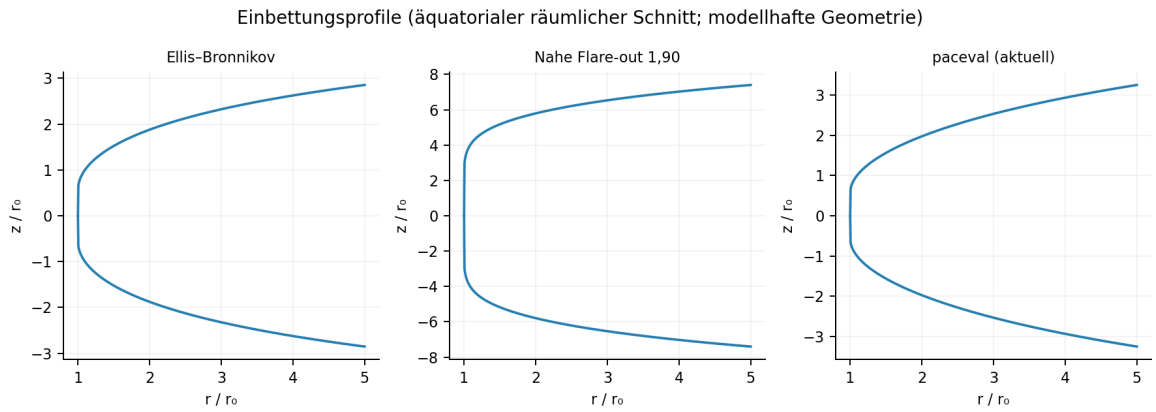



Figure 2. Illustrative spatial-slice embedding diagrams; these do not represent complete four-dimensional spacetimes.

## 5. Numerical results and physical trade-offs

| Model | J_NEC ↓ | Max. radial \|R\| ↓ | Max. K ↓ | Proper distance |
|---|---|---|---|---|
| Ellis special case | 0.990012 | 0.000 | 12.000 | 99.995 |
| Ellis, Φ = −0.75/r | 0.263558 | 0.750 | 14.250 | 99.995 |
| U = 1.90 | 1.446149 | 0.000 | 4.020 | 102.237 |
| U = 1.99 | 1.467755 | 0.000 | 4.000 | 103.613 |
| paceval candidate | 0.103540 | 1.993504 | 27.896 | 100.277 |

Using the same integration protocol, the reduction relative to the massless Ellis special case is (0.990012 − 0.103540)/0.990012 ≈ 89.5%. Relative to the Ellis-type redshift variant, it is approximately 60.7%. This is evidence of a lower radially weighted energy-condition-violation objective, not of a quantitatively established mass or energy saving in a hypothetical construction.

The larger maximum sampled radial tidal curvature (1.9935) and larger Kretschmann scalar (27.896) represent trade-offs. Static orthonormal Riemann components are local curvature indicators; actual tidal accelerations experienced by a traveler would also depend on trajectory, velocity, physical size and the throat's dimensional scale.

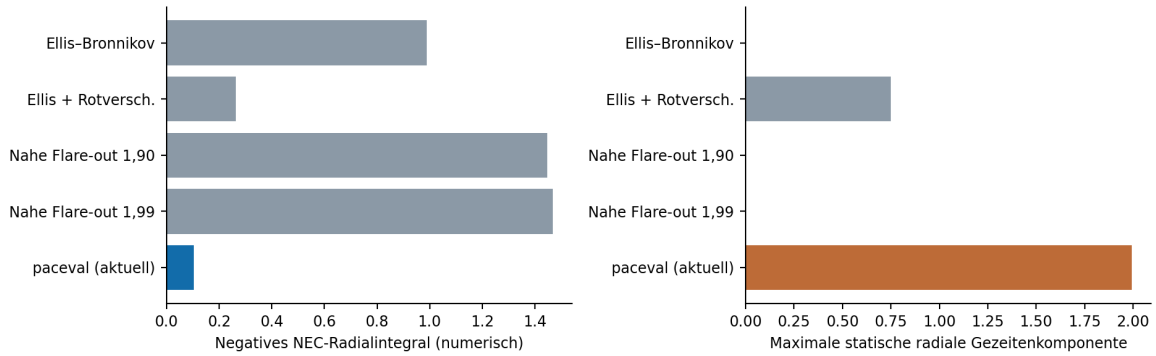



Figure 3. Numerical comparison of the implemented NEC objective and selected curvature indicators. The objective does not measure the absolute amount of required exotic matter.

## 6. Bi-objective search and preliminary Pareto analysis

The search treats two objectives separately: J_NEC and the maximum magnitude of selected static orthonormal tidal Riemann components. Candidate A dominates B if it is no worse in either objective and strictly better in at least one. Nondominance and diversity-based selection are inspired by the established NSGA-II approach [4].

The recorded run archived 100 candidates. For computational reasons, only 24 were re-evaluated on the 1,200-point fine grid. Nineteen of these 24 were nondominated within that subset. They do not constitute 19 globally optimal wormholes or a fully verified front over the full archive. The fine-grid NEC-favorable option has (J_NEC, tidal) = (0.103540, 1.993504); one low-tidal alternative has (1.166020, 0.002347).

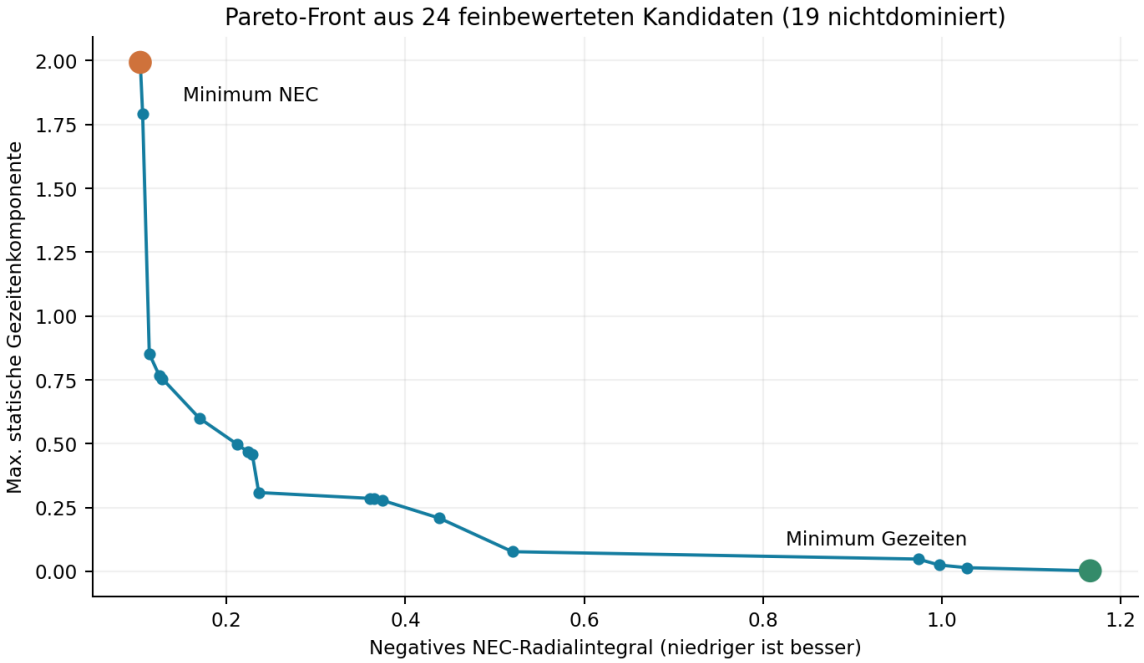



Figure 4. Preliminary Pareto comparison within the evaluated candidate subset. Points denote numerical models, not experimental measurements.

## 7. Reproducibility and numerical verification status

The documented Stage-8 run uses seed 20261029, population 20, 100 candidate evaluations, 72 coarse and 1,200 fine radial samples, and adaptive TINC batch refinement with 256 added points. Three analytical reference cases passed the prescribed checks for twelve computed fields. For the selected candidate, 1,024 regular and 273 adaptive TINC sample points were successfully evaluated. At the regular points, 142 NEC intervals were classified negative and 882 nonnegative, with none inconclusive.

At one radius, the original calculation graph returned paceval error code 144 for four quantities. A separately constructed, algebraically equivalent calculation graph subsequently produced successful pointwise results for all four (attempted 4, recovered 4, unresolved 0). This resolves the particular evaluated point for this run, but does not identify or fix the underlying engine condition that generated code 144. Pointwise TINC success is not a proof over the continuous radial domain.

| Setting | Recorded value |
|---|---|
| Experiment | Stage-8; C++17; paceval SDK 4.25 |
| Random seed / candidate count | 20261029 / 100 evaluations |
| Grids | 72 coarse / 1,200 fine / 1,024 regular TINC points |
| Adaptive TINC | 17 initial + 256 refinements = 273 |
| Pareto fine-grid reassessment | 24 of 100 archived candidates |
| Limitation | No proven TINC bounds over continuous intervals |

## 8. Research context, scientific interpretation and next steps

The basic conditions for traversable Morris–Thorne wormholes have been established since 1988 [1]; the Ellis framework is older [3]. In particular, Visser, Kar and Dadhich showed in 2003 that certain theoretical wormhole geometries can be constructed with arbitrarily small amounts of exotic matter under a specifically defined integral quantifier [2]. It would therefore be inappropriate to infer a worldwide record or a first-ever reduction in exotic matter from the present comparison.

The present contribution is the exploratory integration of symbolic expression search, geometric diagnostics, bi-objective selection and pointwise numerical error checks in an executable prototype. Whether this approach is novel or more effective than established optimization workflows remains an open question requiring controlled benchmarking and independent replication.

### Prioritized follow-up investigations

First: independently derive and verify all tensor expressions, including Riemann sign conventions. Second: re-evaluate the full candidate archive on the fine grid and repeat the search using independent seeds. Third: establish validated error bounds for the integrals and over the continuous radial domain. Fourth: compare with established parametrized wormhole families under identical objectives and constraints. Fifth: study the physical stress-energy source, energy conditions, stability, throat scaling and traveler-dependent tidal accelerations. Sixth: benchmark runtime and energy consumption against independent implementations at comparable accuracy.

## 9. Conclusion

The numerical research prototype demonstrates that static, spherically symmetric wormhole geometries can be generated automatically as symbolic expression trees and assessed against multiple physically motivated objectives. The documented candidate markedly reduces the explicitly defined radial NEC objective relative to the selected Ellis special case, while increasing static tidal curvature. The preliminary Pareto assessment makes the competing geometric objectives transparent.

The results motivate further research into automated mathematical model generation. They do not establish the discovery of an observed wormhole, an engineering-feasible construction, or a literature-wide minimum exotic-matter requirement. The scientifically decisive next step is independent replication supported by continuous-domain validation.

## References and provenance

[1] Morris, M. S.; Thorne, K. S. (1988). Wormholes in spacetime and their use for interstellar travel. American Journal of Physics 56, 395–412. DOI: 10.1119/1.15620.

[2] Visser, M.; Kar, S.; Dadhich, N. (2003). Traversable Wormholes with Arbitrarily Small Energy Condition Violations. Physical Review Letters 90, 201102. DOI: 10.1103/PhysRevLett.90.201102.

[3] Ellis, H. G. (1973). Ether flow through a drainhole: A particle model in general relativity. Journal of Mathematical Physics 14, 104–118. DOI: 10.1063/1.1666161.

[4] Deb, K.; Pratap, A.; Agarwal, S.; Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation 6, 182–197. DOI: 10.1109/4235.996017.

Primary data: recorded Stage-8 run of 30 October 2026; C++17 source wormhole_100_stage8_tinc144_algebraic_retry.cpp. Figures derive from model visualizations previously prepared from these numerical results. This manuscript has not undergone independent peer review.